

Practical week-4

Create Database

The screenshot shows the SQL Developer interface with the 'Query 1' window open. The SQL script contains two statements: `create database pweek4;` and `use pweek4;`. The 'Output' window at the bottom displays the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec

The 'Navigator' pane on the left shows the 'SCHEMAS' tree with 'p_week1' selected. The 'Administration' and 'Information' tabs are visible at the bottom.

Create Books Table (DDL + Constraints)

The screenshot shows the SQL Developer interface with the 'Query 1' window open. The SQL script contains the following DDL statement:

```
CREATE TABLE Books (  
    book_id INT AUTO_INCREMENT PRIMARY KEY,  
    title VARCHAR(200) NOT NULL,  
    author VARCHAR(150) NOT NULL,  
    genre VARCHAR(100),  
    description TEXT,  
    published_year INT  
);
```

The 'Output' window at the bottom displays the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCREME...	0 row(s) affected	0.031 sec

The 'Navigator' pane on the left shows the 'SCHEMAS' tree with 'p_week1' selected. The 'Administration' and 'Information' tabs are visible at the bottom.

Describe Books Table

The screenshot shows a SQL IDE interface with a query editor, a schema navigator, and an output window. The query editor contains the following SQL code:

```
3 CREATE TABLE Books
4 (
5     book_id INT AUTO_INCREMENT PRIMARY KEY,
6
7     title VARCHAR(200) NOT NULL,
8
9     author VARCHAR(150) NOT NULL,
10 )
```

The output window displays the 'Result Grid' for the 'Describe Books' query. The table structure is as follows:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

The 'Output' window shows the 'Action Output' for the query, listing the actions performed and their results:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCREM...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec

Insert 10 Books and Display

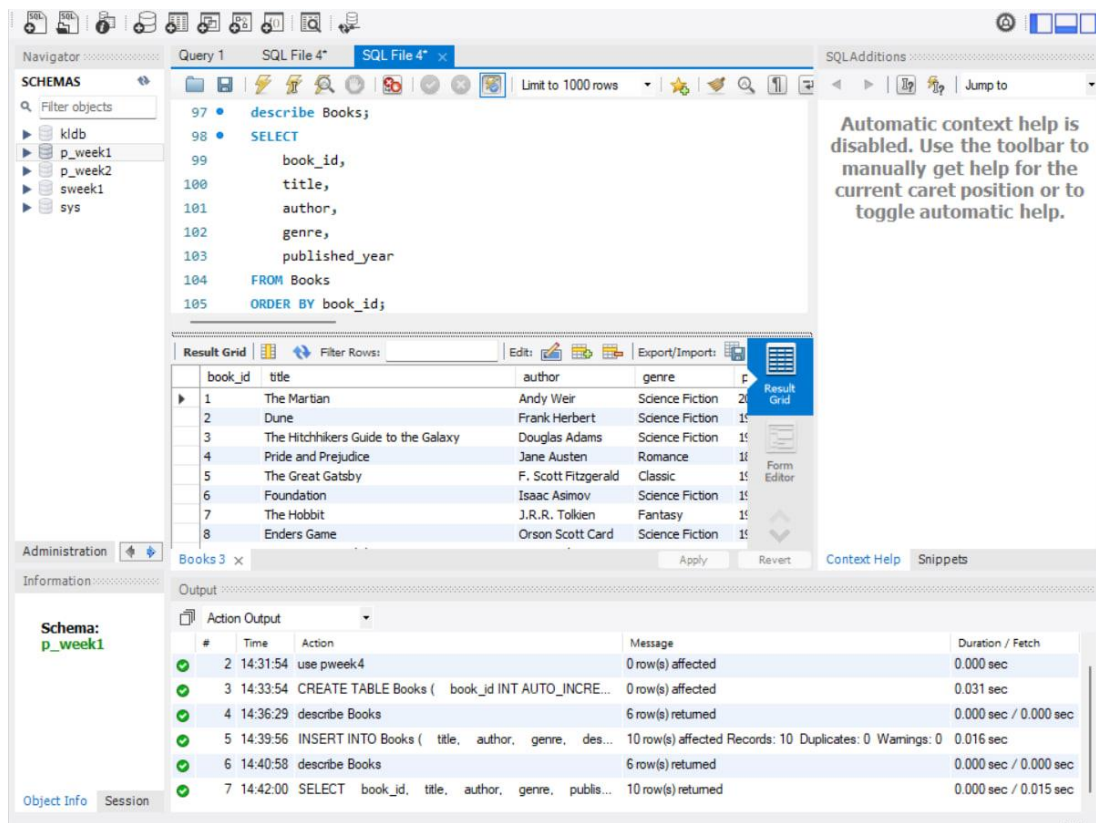
The screenshot shows a SQL IDE interface with a query editor, a schema navigator, and an output window. The query editor contains the following SQL code:

```
16 );
17 describe Books;
18 INSERT INTO Books
19 (
20     title,
21     author,
22     genre,
23     description,
24     published_year
25 )
26 VALUES
27 (
28     'The Martian',
29     'Andy Weir',
30     'Science Fiction',
31     'An astronaut struggles to survive alone on Mars.',
32     2011
33 ),
34 )
```

The output window displays the 'Action Output' for the query, listing the actions performed and their results:

#	Time	Action	Message	Duration / Fetch
1	14:31:10	create database pweek4	1 row(s) affected	0.016 sec
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCREM...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
5	14:39:56	INSERT INTO Books (title, author, genre, descr...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec

Add Embedding Column (JSON)



Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
97 • describe Books;
98 • SELECT
99     book_id,
100     title,
101     author,
102     genre,
103     published_year
104 FROM Books
105 ORDER BY book_id;
```

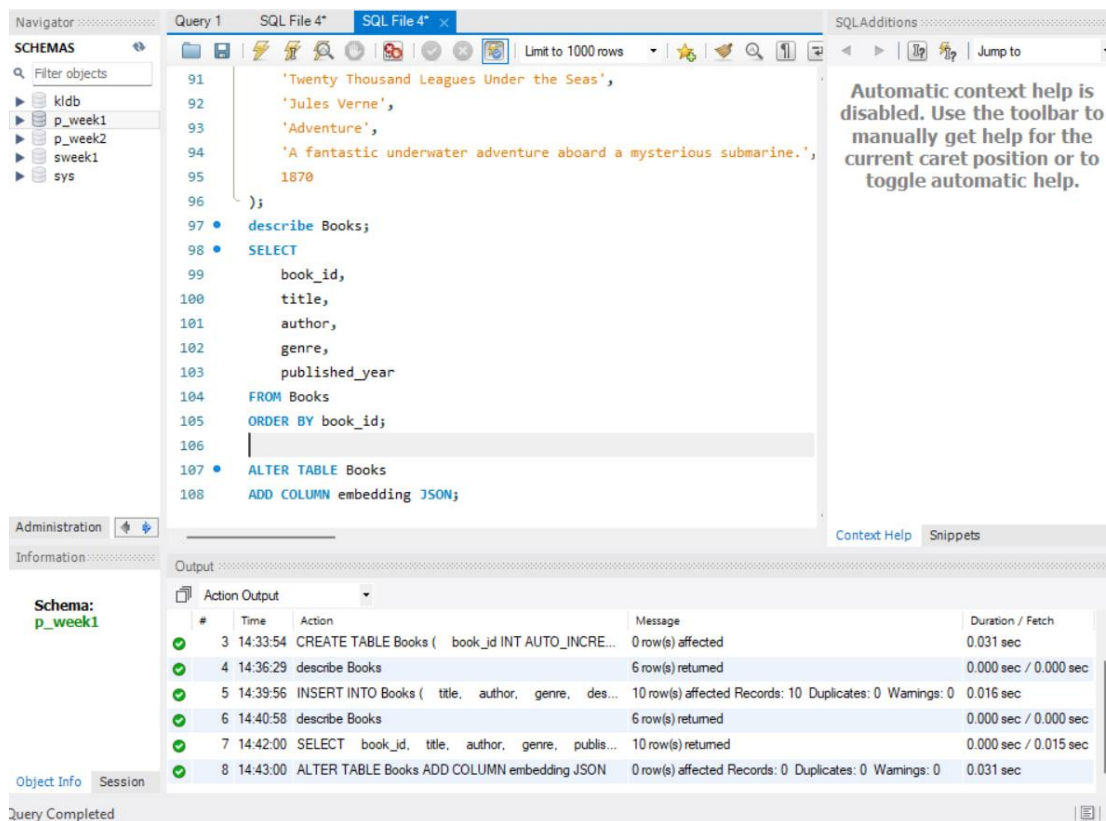
book_id	title	author	genre
1	The Martian	Andy Weir	Science Fiction
2	Dune	Frank Herbert	Science Fiction
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction
4	Pride and Prejudice	Jane Austen	Romance
5	The Great Gatsby	F. Scott Fitzgerald	Classic
6	Foundation	Isaac Asimov	Science Fiction
7	The Hobbit	J.R.R. Tolkien	Fantasy
8	Enders Game	Orson Scott Card	Science Fiction

Schema: p_week1

Output: Action Output

#	Time	Action	Message	Duration / Fetch
2	14:31:54	use pweek4	0 row(s) affected	0.000 sec
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCRE...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
5	14:39:56	INSERT INTO Books (title, author, genre, des...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
6	14:40:58	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
7	14:42:00	SELECT book_id, title, author, genre, publis...	10 row(s) returned	0.000 sec / 0.015 sec

Update Book Embeddings



Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
91     'Twenty Thousand Leagues Under the Seas',
92     'Jules Verne',
93     'Adventure',
94     'A fantastic underwater adventure aboard a mysterious submarine.',
95     1870
96 );
97 • describe Books;
98 • SELECT
99     book_id,
100     title,
101     author,
102     genre,
103     published_year
104 FROM Books
105 ORDER BY book_id;
106
107 • ALTER TABLE Books
108 ADD COLUMN embedding JSON;
```

Schema: p_week1

Output: Action Output

#	Time	Action	Message	Duration / Fetch
3	14:33:54	CREATE TABLE Books (book_id INT AUTO_INCRE...	0 row(s) affected	0.031 sec
4	14:36:29	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
5	14:39:56	INSERT INTO Books (title, author, genre, des...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec
6	14:40:58	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
7	14:42:00	SELECT book_id, title, author, genre, publis...	10 row(s) returned	0.000 sec / 0.015 sec
8	14:43:00	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

Query Completed

Display Book Embeddings

The screenshot shows a SQL Developer window with a query editor containing a series of UPDATE statements for a table named 'Books'. The queries are numbered 122 through 140. Each query updates the 'embedding' column with a JSON_ARRAY of three values. The 'WHERE' clause for each query specifies a unique 'book_id' from 4 to 10. The interface includes a 'Navigator' pane on the left showing a schema named 'p_week1'. The 'Output' pane at the bottom shows the execution results of these queries, indicating that each update affected one row and took 0.000 seconds. A message box on the right states: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

```
122 WHERE book_id = 4;
123 UPDATE Books
124 SET embedding = JSON_ARRAY(0.15, 0.30, 0.80)
125 WHERE book_id = 5;
126 UPDATE Books
127 SET embedding = JSON_ARRAY(0.95, 0.65, 0.20)
128 WHERE book_id = 6;
129 UPDATE Books
130 SET embedding = JSON_ARRAY(0.30, 0.75, 0.90)
131 WHERE book_id = 7;
132 UPDATE Books
133 SET embedding = JSON_ARRAY(0.88, 0.85, 0.15)
134 WHERE book_id = 8;
135 UPDATE Books
136 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
137 WHERE book_id = 9;
138 UPDATE Books
139 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
140 WHERE book_id = 10;
```

#	Time	Action	Message	Duration / Fetch
15	14:46:59	UPDATE Books SET embedding = JSON_ARRAY(0.15, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
16	14:47:16	UPDATE Books SET embedding = JSON_ARRAY(0.95, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
17	14:47:30	UPDATE Books SET embedding = JSON_ARRAY(0.30, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
18	14:47:44	UPDATE Books SET embedding = JSON_ARRAY(0.88, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.015 sec
19	14:47:57	UPDATE Books SET embedding = JSON_ARRAY(0.20, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec
20	14:48:16	UPDATE Books SET embedding = JSON_ARRAY(0.70, ...	1 row(s) affected Rows matched: 1 Changed: 1 Wamin...	0.000 sec

L2 Distance Query

The screenshot shows a SQL Developer window with a query editor containing a SELECT statement. The query selects 'book_id', 'title', and 'embedding' from the 'Books' table, ordered by 'book_id'. The interface includes a 'Navigator' pane on the left showing a schema named 'p_week1'. The 'Output' pane at the bottom shows the execution results of this query, indicating that 10 rows were returned and it took 0.000 seconds. A message box on the right states: 'Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.'

```
140 WHERE book_id = 10;
141
142 SELECT
143     book_id,
144     title,
145     embedding
146 FROM Books
147 ORDER BY book_id;
```

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]

Cosine Similarity Query

The image shows two screenshots of a database management tool interface, likely DBeaver, demonstrating SQL queries for cosine similarity calculation.

Top Screenshot: Query 1

Query 1 SQL:

```
SELECT
    book_id,
    title,
    author,
    genre,
    ROUND(
        SQRT(
            POW(
```

Result Grid:

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	Dune	Frank Herbert	Science Fiction	0.0866

Bottom Screenshot: Query 6

Query 6 SQL:

```
genre,
ROUND(
(
    CAST(
        JSON_EXTRACT(embedding, '$[0]')
        AS DECIMAL(10,6)
    ) * 0.90
    +
    CAST(
```

Result Grid:

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9986
8	Enders Game	Orson Scott Card	Science Fiction	0.9983

The interface includes a Navigator panel on the left showing schemas (kldb, p_week1, p_week2, sweek1, sys) and a Schema: p_week1. The bottom panel shows the Action Output log with timestamps and messages for each query execution.